

SHOAIB MOHAMMAD

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Education

The Ohio State University

Masters, Computer Science and Engineering

2024-2026

CGPA - 3.64/4.0

Indian Institute of Technology, Kharagpur

B.Tech, Micro spl. in Artificial Intelligence and Applications, Major in Civil Engineering

2020-2024

CGPA - 8.4/10

VelocIITy, junior college

High school, class 11 and 12

2017-2019

96.9%

Geethanjali high school, class 8 - 10

2015-2017

Research and Industrial Experience

MPI Communication Performance on AMD MI300A: Microbenchmarks and Applications

Nov 2025

Co-authored a research paper to publish in International Conference for High Performance Computing workshop(SC' 25)

- Presented a comparative study of *MVAPICH-Plus*, *Open MPI*, *MPICH*, *Cray MPICH* on 1-32 **MI300A** APU nodes
- Built and set up NOWLAB's PyTorch 2.7 and a GPT2 model with various MPI stacks for **DDP training** performance
- Measured per step iteration time of DDP training and shown that MVAPICH-PIUS 4.1 and CRAYMPICH outperformed Open MPI by more than **99%** and MPICH by more than **93%** at 32 nodes with altogether 128 APUs

NOWLAB - OSU | Graduate Research Associate—Professor Dhabaleswar K Panda[Dec 2024 - Present]

Goal: Enhancing PyTorch library for Distributed Training of Deep Learning Models with MPI

- Implemented **reduce-scatter,all-gather-base** algorithms to support FSDP with MPI in NOWLAB's PyTorch repo
- Rebased and tested NOWLAB's pytorch repo with PyTorch 2.7.0 and 2.8.0 for distributed training with Lab members
- Evaluated **DDP per step training** with a GPT2 model using latest allreduce designs of MVAPICH-Plus 4.1
- Assisted in the release of **Hidl v1.0 and v2.0** by generating wheels on AMD MI300, NVIDIA GB200, H100 HPC clusters, and evaluating the DDP training performance up to **512** GPUs

PARENTHESSES LABS | Software Engineer Intern

May 2023 - Jul 2023

- Analyzed **1M+** data, designed schema, connected to an **API** endpoint of a cloud database with aid of **psql** commands
- Generated **40+** graphs using **Metabase SQL**, interface queries and creation of complex schema to map attributes
- Deployed the local metabase web app instance in **Micorsoft Azure** using azure DevOps organizations and Docker

Projects

EdgeFlexx-Transformer: Transformer Inference for Edge Devices

Jan 2025 - Apr 2025

Research course Project — Computer Networking, CSE, OSU

- Co-designed model execution with memory management for **adaptive inference framework** for edge devices
- Developed a pipeline integrating **Memory-aware** model pruning, **Activation-aware quantization**, and mixed precision support using context-aware prediction and pre-profiling to decrease memory requirement and latency
- Optimized memory usage by **50%** and latency by **83%** with model pruning and FP16 while maintaing accuracy
- Reduced peak memory consumption by **75%** with Activation aware quantized and pruned transformer model

LLMs-Stack

Sep 2024 - Nov 2024

Research course Project — Adv AI, CSE, OSU

- Designed a **memory-efficient LLMs ensemble framework** that selects specialized models based on input modality
- Optimized memory usage by a **35.4%** reduction in average memory requirements, maintaining accuracy across tasks
- Developed a novel **dynamic contrastive reasoning(DCR)** with pruned models to enhance reasoning capabilities
- Evaluated performance using semantic similarity, ensuring accurate assessment of both answers and reasoning outputs

Unmasking adversarial usage of GenAI in Online hate

Aug 2023 - Nov 2023

Research Course Project — Prof. Mainack Mondal, CSE dept, IIT Kharagpur

- Designed system and built a model that successfully decreased the efficiency by **20%** of a Facebook's hugging face roberta-hate-speech-dynabench-r4-target online hate detection model
- Developed a **pipeline** integrating **KeyBERT**, Roberta model, and perturbation technique to rewrite the input text
- Validated the results with a survey that gave **65-70%** validity among all perturbed texts generated by our model

Relevant Coursework and Certifications

- High-Performance Deep Learning
- Foundations of Programming Languages
- Intermediate Computer Networking
- Algorithms
- Advance Operating System
- Advance Computer Architecture
- Advance Artificial Intelligence
- Graphical and Generative Modeling of Machine Learning
- Machine Learning Foundations and Applications
- **Natural Language Processing Specialization** | *DeepLearning.AI* | Coursera
- **Machine Learning Specialization** | *DeepLearning.AI and Stanford University* | Coursera

Competitions

ChatWithMe, Hack-I/O, The Ohio State University, CSE Hackathon Nov 2025

- Built a **Personal Assistant** AI-Agent for content creators that manages user queries and helps in affiliate marketing
- Integrated Grok AI with RAG flow from Supabase for real-time updated context to respond with accuracy
- Created database management pipeline using Supabase for every content creator and voice-enabled assistant

VITALS MD, The Ohio State University CSE + Medical Hackathon Nov 2024

- Developed a product that records our vitals using Medical Image Analysis with OpenAI API for computer vision
- Built a dashboard that has visual information and an analytics dashboard of our vitals rendering using Streamlit

Stripe BigCommerce Global hackathon Apr 2025

- Awarded with the **BigDesign** award, also known as an honorable mention in Stripe x BigCommerce global hackathon among 80+ teams that participated throughout the world
- Innovated and developed a new product by fusing the BigCommerce service technology with the Stripe terminal SDK

Technical Skills

Programing languages and libraries: C, C++, Python, Javascript, SQL, PyTorch, Tensorflow

Softwares and Frameworks : Linux, NextJs, ExpressJs, PostgreSQL, GraphQL, MongoDB, Matlab, AutoCAD

Awards and Achievements

- Won **Bronze** in the interdisciplinary OSUCOM Appathon of CSE and Medical college for novel use of AI in Medicine
- Secured **99.57** and **99.04** percentiles in IIT-JEE Advanced 2020 and in JEE Mains 2020 among 1.2M+ candidates.
- Awarded with the **BigDesign** award, in Stripe x BigCommerce global hackathon among 80+ participating teams

Extracurricular Activities

Treasurer | *Civil Engineering Society, IIT Kharagpur* Apr 2023-May 2024

- Orchestrated **17 events, logistics** and **800k+ budget** in a team for smooth functioning of the fest, **Megalith 2023**
- Primed the workflow, vision, timeline, etc of Megalith 2024 to the whole team, initiated new activities in the fest
- Planned and volunteered **16+ social** events for students and professors of the Civil Department, managed **150k+ budget**

Secretary | *Social and Cultural Events | Azad Hall of Residence, IIT Kharagpur* Dec 2021-Apr 2022

- Led the **Fine Arts** teams to inter-hall **Gold** and **Bronze** medals by having well-organized and scheduled practices
- Managed the necessary **logistics** and budget of **200k+** by handling **30+** events for **700+** residents of Azad Hall

Mentor | *Student Welfare group, IIT Kharagpur* Nov 2022-Mar 2024

- Student Mentor of the Civil Engineering Society for the academic session of 2022-2023 to guide 2nd-year students

Others

- Served as a **National Service Scheme** student for 2 years at IIT Kharagpur by designing presentations and documentations for more than 10 government schemes in a simple way that aids the poor and illiterate people.
- Part of the **silver-winning team** in choreography in the inter-hall competition held at IIT Kharagpur in March 2022
- Captained a high school team in 2016, **achieved 2nd** position by conducting assembly and weekly activities, games
- **Wing Representative** of Azad Hall of Residence, E-Block east wing, IIT Kharagpur for the academic year 2022-2023